

SIMULTANEOUS IDENTIFICATION OF INFLUENTIAL SUBSETS AND VARIABLE SELECTION USING QUANTABILITIES

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Key words: Dummy variables; holistic regression; influential observations; mean-shift outlier model; outlier detection; quantabilities; variable selection.

Summary: Ideally, outlier detection and variable selection should be dealt with at the same time. Literature, however, proposes only a few practical procedures for handling this problem. A procedure is proposed where dummy variables, variable selection, and a simple selection rule are used to find an optimal model which includes the relevant predictors and excludes the outliers influential on the fit.

1. Introduction

Many techniques have been proposed for the separate selection of variables and the identification of outliers, but relatively little has been done to address these problems simultaneously. For a review of the individual approaches, see Hocking (1976) in respect of the variable selection problem, and Chatterjee and Hadi (1986) in respect of the outlier problem.

Most approaches to the combined problem consist of a two-step approach where predictors are first selected, and the outliers subsequently detected, or vice versa. Such a sequential procedure “can be regarded as analogous to one-variable-at-a-time methods and has similar disadvantages” (Peixoto and LaMotte, 1989).

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Reasons for the simultaneous identification of influential subsets and variable selection are:

- Predictors can appear to be important (or not) because of just a few influential observations or outliers.
- Different subsets of predictors can display completely different influence and leverage patterns for the same response variable.
- One-at-a-time case diagnostics are not suitable for situations where cases are jointly (but not individually) outlying or influential.
- Finally the influence of some observations only become apparent once others have been left out (the problem of masking).

We hence support Peixoto and LaMotte (1989), Ahn and Park (1987), Hoeting, Raftery and Madigan (1996), and Allen (1984) who all agree that (in the words of the latter)

“it seems imperative that any scheme to deal with high leverage and outliers must simultaneously deal with variable selection”.

Several authors address the problem of simultaneous variable selection and outlier detection. Ahn and Park (1987) propose a weighted sum of mean squared errors with case weighting as a criterion for variable selection, while Hoeting, Raftery and Madigan (1996) use a Bayesian approach to propose a method based on the computation of posterior model probabilities. Peixoto and LaMotte (1989) utilise dummy variables for all cases as an extension of the mean-shift outlier model and point out the problems associated with the overdefined model, as well as the unsatisfactory results of some of the available variable selection procedures. They, however, fail to come up with a general selection rule, which is crucial, especially in the case where all possible subsets are to be evaluated. We adopt the same approach as Peixoto and LaMotte and

in addition propose a simple, but effective selection rule that can be used to select predictors simultaneously and identify the observations influential on the selection.

2. The mean-shift outlier model

Consider the standard linear model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \quad (1)$$

where $\mathbf{y}$ is an $n \times 1$ vector of observed values of the response variable, $\mathbf{X}$ is an $n \times p$ matrix of rank p , $\boldsymbol{\beta}$ is a $p \times 1$ vector of unknown parameters, and $\boldsymbol{\epsilon}$ is an $n \times 1$ vector of random errors with mean $\mathbf{0}$ and covariance $\sigma^2 \mathbf{I}_n$.

The least squares estimator of $\boldsymbol{\beta}$ is given by $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$, the residual mean square

$$\begin{aligned} \hat{\sigma}^2 &= \frac{\mathbf{y}'(\mathbf{I} - \mathbf{H})\mathbf{y}}{n - p} \\ &= \frac{\mathbf{e}'\mathbf{e}}{n - p} \end{aligned} \quad (2)$$

is an estimator of σ^2 and $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is the hat matrix with diagonals h_{ii} . The dummy variable d_i is equal to one for case i and equal to zero for all other cases. The i th residual is denoted by e_i and the sample variances of y and X_j are denoted by S_y^2 and $S_{X_j}^2$, respectively. The subscript i is used to identify the i th case and the subscript j is used to identify the j th predictor. Let $\mathbf{y}_{(-i)}$, $\mathbf{X}_{(-i)}$ and $\hat{\sigma}_{(-i)}^2$ denote $\mathbf{y}$, $\mathbf{X}$ and $\hat{\sigma}^2$ with the i th case deleted, while $\hat{\sigma}_{[-j]}^2$ denotes $\hat{\sigma}^2$ with the j th predictor omitted. Introducing the j th predictor into the model is denoted by $[+j]$, so that for example the residual sum of squares SSE becomes $\text{SSE}_{[+j]}$ and the coefficient of determination R^2 becomes $R_{[+j]}^2$.

The mean-shift outlier model is defined as

$$\mathbf{y} = \mathbf{X}\beta + \theta \mathbf{d}_i + \epsilon \quad (3)$$

where $\mathbf{d}_i$ is a vector with 1 in the i th position, and zeros elsewhere. It can be shown that the least squares estimator of θ is given by

$$\begin{aligned} \hat{\theta} &= \frac{e_i}{1 - h_{ii}} \\ &= e_{(-i)} \end{aligned} \quad (4)$$

which is the i th predicted residual from the model

$$\mathbf{y}_{(-i)} = \mathbf{X}_{(-i)}\beta + \epsilon_{(-i)}. \quad (5)$$

These two models, (3) and (5), are related and although they have different values for the total (corrected) sum of squares and related statistics, they share the same residual sum of squares, residual mean square and least squares estimate of β .

Furthermore

$$\text{var}(\hat{\theta}) = \frac{\sigma^2}{(1 - h_{ii})} \quad (6)$$

so that

$$\begin{aligned} t_i &= \frac{\hat{\theta}}{\hat{\sigma}_{(-i)}(1 - h_{ii})^{-\frac{1}{2}}} \\ &= \frac{e_i}{\hat{\sigma}_{(-i)}(1 - h_{ii})^{\frac{1}{2}}} \end{aligned} \quad (7)$$

is the i th studentised residual, where $t_i \sim t_{n-p-1}$ under the conditions that $\text{rank}(\mathbf{X}_{(-i)}) = p$ and $\epsilon \sim N(\mathbf{0}; \sigma^2 \mathbf{I}_n)$. Hence the t statistic for $H_0: \theta = 0$ is identical to the test for an outlier. Here 1 degree of freedom is used to fit the dummy variable d_i so that $h_{ii} = 1$ and $\hat{y}_i = y_i$. Note that in a t -directed search the t_i for the i th dummy variable and the t_j for the j th predictor are in direct competition in the sense that (depending on the size of the t values)

omitting the i th observation can lead to a greater improvement in the model than including the j th predictor.

3. The holistic regression model

Following Peixoto and LaMotte (1989), we define the most general or holistic model as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{I}_n\boldsymbol{\theta} + \epsilon \quad (8)$$

where

$$\mathbf{I}_n\boldsymbol{\theta} = \theta_1\mathbf{d}_1 + \theta_2\mathbf{d}_2 + \theta_3\mathbf{d}_3 + \dots + \theta_n\mathbf{d}_n \quad (9)$$

This model is overdefined, that is, it has $n + p$ parameters versus only n observations, and consequently the full model cannot be fitted. This implies that no stopping rule based on the error sum of squares for the full model, for example C_p , can be utilized, since SSE approaches zero as the degrees of freedom for the model approach $n - 1$. In fact, this phenomenon makes all the popular selection criteria redundant under these circumstances, since they are all functions of the error sum of squares (SSE_k) for the k -term fitted model.

They are amongst others the mean squared error (MSE_k), the final prediction error (J_k), the prediction mean squared error (S_k), Akaike's information criterion (AIC), Sawa's Bayesian information criterion (BIC), the estimated mean squared error of prediction (GMSEP), Amemiya's prediction criterion (PC), and Schwarz's Bayesian criterion (SBC).

Consider a typical stepwise variable selection procedure for the model given in (8). Note that such a procedure entails a minimum of $n - 1$ steps (if no stopping rule is used), each involving a candidate model. The process uses an improvement criterion (usually related to the largest increase in R^2). With each new inclusion of a variable, a degree of freedom is lost so that after a (minimum) of $n - 1$ steps, zero degrees of freedom remain and the error

sum of squares is zero. It is this phenomenon that causes the above-mentioned selection criteria to break down.

The dummy variable model needs some special clarification. The first variable to enter the model is the one which has the highest (absolute) correlation with y . It readily follows that the correlation between the dummy d_i and y is given by

$$r_{d_i, y} = \frac{\sqrt{n}(y_i - \bar{y})}{(n-1)S_y} \quad (10)$$

where $S_y = \sqrt{\sum_i^n (y_i - \bar{y})^2 / (n-1)}$ is the sample standard deviation of y .

Let, at a particular stage of the model building process, SSE and R^2 denote the residual sum of squares and the coefficient of multiple determination respectively. The following variable to enter the model is one that has the largest partial sum of squares (extra sum of squares, type II sum of squares) say L . For X_j , L_j is given by

$$\begin{aligned} L_j &= \text{SSE} - \text{SSE}_{[\dagger j]} \\ &= \hat{\beta}_j^2 \cdot \text{SSE}_{X_j} \end{aligned} \quad (11)$$

where SSE_{X_j} is the residual sum of squares when X_j is regressed on the predictors already in the model. The partial sum of squares for d_i is given by

$$\begin{aligned} L_i &= \text{SSE} - \text{SSE}_{(-i)} \\ &= \frac{e_i^2}{(1 - h_{ii})} \end{aligned} \quad (12)$$

If $L_i > L_j$, omitting the i th observation (including the i th dummy variable) leads to a greater improvement in the model than including the j th predictor.

Another approach to understanding the selection procedure is to convert L to a (squared) semi-partial correlation δ^2 by dividing L by the total (corrected) sum of squares for y . Hence

$$\begin{aligned}
 \delta_j^2 &= \frac{L_j}{(n-1)S_y^2} \\
 &= \frac{\hat{\beta}_j^2 \cdot \text{SSE}_{X_j}}{(n-1)S_y^2} \\
 &= \frac{\hat{\beta}_j^2 \cdot (S_{X_j}^2 / S_y^2)}{(n-1)S_{X_j}^2 / \text{SSE}_{X_j}} \\
 &= \frac{\hat{\beta}_j^{*2}}{\text{VIF}_j}
 \end{aligned} \tag{13}$$

Here $\hat{\beta}_j^*$ is the standardized parameter estimate of β_j , VIF_j is the variance inflation factor for X_j , and S_y^2 is the variance of y . This is a typical signal-to-noise ratio and it is clear from (13) that a large increase in R^2 can be brought about by large values of the (squared) standardised coefficients (signal), but a predictor with a large variance inflation factor (noise) will counteract this effect.

For the dummy variable d_i ,

$$\begin{aligned}
 \delta_i^2 &= \frac{L_i}{(n-1)S_y^2} \\
 &= \frac{e_i^2}{(n-1)S_y^2(1-h_{ii})} \\
 &= \frac{e_i^2 / (nS_y^2(1-h_{ii})^2)}{((n-1)/n)(1/(1-h_{ii}))} \\
 &= \frac{\hat{\theta}_i^{*2}}{\text{VIF}_i}
 \end{aligned} \tag{14}$$

where $\hat{\theta}_i^*$ is the standardised parameter estimate of θ_i , VIF_i is the variance inflation factor for d_i , and S_y^2 is the variance of y . Since the (relative) size of e_i^2 and the size of the leverage h_{ii} are in contrast, it follows from (14) that omitting a low leverage outlier (e_i^2 large, h_{ii} small) will substantially increase δ_i^2 , but a high leverage observation (h_{ii} large, e_i^2 small) will deter the increase and hence resist the removal of this particular observation.

4. Finding the optimal subset

The correlation matrix $C : (k + 1) \times (k + 1)$ (of all correlations between the predictors and the dependent variable) plays an important part when it comes to variable selection, since subset models can be evaluated without actually fitting the models. For k predictors, R_k^2 can be efficiently calculated as

$$1 - R_k^2 = \frac{|C|}{|C_k|} \quad (15)$$

where $|C|$ is the determinant of C and $|C_k|$ is the determinant of the submatrix C_k $k \times k$ corresponding to the k predictors. Hence for a model with n observations and k predictors, the F -statistic can be computed as

$$F = \frac{R_k^2 \cdot (n - k - 1)}{(1 - R_k^2) \cdot k} \quad (16)$$

from which the true significance level can be calculated. Transforming C to upper triangular form (using Gaussian elimination) further reduces computational costs, since the determinants then become products of the appropriate diagonals (see Drezner, Marcoulides and Salhi, 1999).

For a fixed value of k , define the k -variable optimal (best subset) model as the k -variable model that has the smallest observed significance level (tail probability, p-value) among all k -variable models. It is clear that this model will have the largest value R_k^2 and F-statistic among the k -variable models, since they have the same degrees of freedom. Finally define the overall optimal model as the model that has the smallest observed significance level among all best subset models. Within the context of the holistic regression model these probabilities can be very small (as $R^2 \rightarrow 1$) and we refer to them as quantabilities, Q_k (from quantum and probability). The selection procedure therefore is essentially a minimization criterion like all the other popular selection criteria mentioned in this paper. However, unlike the popular selection criteria, quantabilities are not necessarily monotone decreasing, even when the model is overspecified. In most cases it is capable of selecting a subset model of dimension $k \leq n - 1$. This unique property is illustrated by using several data sets.

5. The selection procedure

The selection procedure is fairly simple in that it entails finding the overall optimal model in a progressive way.

This is done by

- fitting all possible subset models and obtaining all k -variable optimal models (for $k = 1, 2, \dots, n - 1$), for which the quantabilities are subsequently calculated;
- obtaining the overall optimal model as the model corresponding to the first entry when sorting the quantabilities in ascending order.

The formulas of Abramowitz and Stegun (1968) can be used for this purpose. The optimal model indicates which predictors should be included in the model, together with the observations (outliers) that should be excluded.

A word of caution should be given here. The above procedure must be strictly observed when evaluating all the best subset models via the emerging pattern in the quantabilities. For most of the many data sets we have analyzed, the quantabilities decrease from the case $k = 1$ to the minimum value for the optimal model and from there on they increase again, hence forming a single trough. However, in some cases with weaker data, the smallest quantability is found close to or at the last model for which $k = n - 1$. This implies that a large number of “outliers” are flagged. A solution for the problem is to select a suboptimal model with a relatively large value of the RTOI (relative-to-optimal index), but which has fewer dummies. Define the relative-to-optimal index as

$$\text{RTOI}_k = \frac{Q_{\text{optimal}}}{Q_k} \cdot 100\%; \quad k = 1, 2, \dots, n - 1.$$

Another observed phenomenon is that a data set has two or more troughs in terms of quantabilities. Under these circumstances the “best choice” for a subset model is not necessarily the one with the smallest quantability (largest RTOI). We recommend that the more simple model at the first trough (highest local RTOI) is to be preferred to models at subsequent troughs where (many) more of the observations have to be discarded.

While the proposed procedure is a useful diagnostic, care should be taken not to use it as an automatic procedure for selecting models. Model selection should be based on the principles of parsimony, high RTOI values and common sense.

The computational cost of implementing this method becomes high if the total number of predictors and dummy variables exceeds 30. Hence, while waiting for the quantum computer, this method is more feasible for smaller data sets. Note that the computational cost can be reduced by using a stepwise method such as the popular maximum R^2 improvement method (with some

risk of not finding the optimal model), or by using a very efficient and very fast stepwise method such as a TABU search (Drezner, Marcoulides and Salhi, 1999). However, since finding the optimal model cannot be guaranteed, these methods are not recommended.

6. Examples

6.1 Wood Beam Data

The wood beam data set was analyzed by Hoaglin and Welsch (1978). The response (Y) is strength of wood beams and the predictors are gravity (X_1) and moisture content (X_2). Since only 10 wood beams are considered, the number of printed statistics is limited. By utilizing 10 dummy variables and fitting all possible subsets (4095), the best subset models for each dimension are evaluated and the results are reported in the following tables.

Table 1: Candidate models for the Wood Beam Data based on Quantabilities

Model	Predictors	Dummies	df1	df2	F	Quantability	RTOI
1	X_1		1	8	40.13	2.24139E-04	5%
2	X_1	D_1	2	7	85.33	1.21405E-05	100%
3	X_1	$D_1 D_6$	3	6	87.29	2.43788E-05	50%
4	X_1	$D_1 D_6 D_{10}$	4	5	102.06	5.58807E-05	22%
5	$X_1 X_2$	$D_1 D_5 D_6$	5	4	156.06	1.13214E-04	11%
6	$X_1 X_2$	$D_1 D_5 D_6 D_{10}$	6	3	454.04	1.59402E-04	7%
7	$X_1 X_2$	$D_2 D_3 D_4 D_5 D_9$	7	2	7985.04	1.25224E-04	10%
FULL	$X_1 X_2$		2	7	31.50	3.16324E-04	4%

It is clear that the general selection criteria are degenerate, while the optimal model (corresponding to the smallest quantability) is the model based on the predictor X_1 and with the first observation (outlier) omitted.

6.2 Hocking-Pendleton Data

This artificial data set (Hocking and Pendleton, 1983) has 26 observations, 3 predictors and a constant term. The data were generated by the model:

$$y = 20 + 3x_1 - 2x_2 + e_1 \quad (17)$$

where $e_1 \sim N(0 ; 0.25)$. The predictor X_3 was created as a linear function of X_1 and X_2 :

$$2x_3 = 60 - 3x_1 - 1.5x_2 + e_2 \quad (18)$$

where $e_2 \sim N(0 ; 0.16)$. Observations 11, 17 and 18 were generated to have large residuals and observation 24 was generated to obscure the collinearity defined by (18).

By fitting the holistic regression model and fitting all possible subsets, the quantabilities were computed for the optimal subsets. The results are shown in table 3.

The true source of the data (17) is exposed as the model based on the predictors X_1 and X_2 and on the reduced data, having omitted observations 11, 17, 18 and 24. In their paper, Hocking and Pendleton investigate no fewer than 10 different methods on which to model the data. These vary from deleting observations, using robust regression and ridge regression methods. Of these only the 10th method (omitting X_3 together with observations 11, 17, 18 and 24) leads to acceptable results. The resulting least squares fit yields the model

$$\hat{y} = 19.483 + 3.032x_1 - 1.999x_2 \quad (19)$$

with $\hat{\sigma}^2 = 0.236$.

Table 3: Candidate models for the Hocking-Pendleton Data

Model	Predictors	Dummies	df1	df2	F	Quantability	RTOI
1	X_2		1	24	30.37	1.14648E-05	< 1%
2	X_1X_2		2	23	143.91	9.92108E-14	< 1%
3	X_1X_2	D_{18}	3	22	150.71	8.12423E-15	< 1%
4	X_1X_3	$D_{18}D_{24}$	4	21	227.05	5.88547E-17	< 1%
5	X_2X_3	$D_{11}D_{17}D_{24}$	5	20	346.88	1.03535E-18	< 1%
6	X_1X_2	$D_{11}D_{17}D_{18}D_{24}$	6	19	788.50	9.93752E-22	100%
7	X_1X_2	$D_6D_{11}D_{17}D_{18}D_{24}$	7	18	826.92	2.97925E-21	33%
8	X_1X_2	$D_6D_{11}D_{17}D_{18}D_{20}D_{24}$	8	17	883.29	1.02436E-20	10%
FULL	$X_1X_2X_3$		3	22	107.61	2.67586E-13	< 1%

6.3 Health Club Data

The data set is discussed by Chatterjee and Hadi (1988). It consists of the records of 30 employees who were regular members of a company's health club. The predictors are:

X_1 (weight in pounds),

X_2 (resting pulse rate per minute),

X_3 (arm and leg strength),

X_4 (time in seconds in a 1/4-mile trial run).

The dependent variable Y is the time (in seconds) in a one mile run. Chatterjee and Hadi note that the only (highly) significant predictors are X_1 and X_4 . Furthermore they identify observation 30 as an outlier, observation 23 as a high-leverage point, and observation 28 as the most influential on $\hat{\beta}$ and $\hat{\sigma}$.

Here model 3 is the one recommended, since it appears at the first trough and has a large RTOI of 85%. Hence predictors X_1 and X_4 should be included in the model, while observation 28 should be left out.

6.4 Scottish Hill Race Data

This data set was introduced by Atkinson (1986) and was supplied by the Scottish Hill Runners Association. The dependent variable is the record time (in minutes) of 35 hill races, and the two predictors are X_1 (distance in miles) and X_2 (elevation in feet). Atkinson concludes that observations 7, 18 and 33 are outliers. However, 33 is masked and can only be detected once 7 and 18 have been removed. These results are confirmed by Hoeting et al. (1996).

Table 4: Candidate models for the Health Club Data

Model	Predictors	Dummies	df1	df2	F	Quantability	RTOI
1	X_4		1	28	71.72	$3.29707\text{E}-09$	1%
2	$X_1 X_4$		2	27	65.03	$4.74487\text{E}-11$	65%
3	$X_1 X_4$	D_{28}	3	26	52.45	$3.65400\text{E}-11$	85%
4	$X_1 X_4$	$D_{21} D_{28}$	4	25	43.79	$6.08120\text{E}-11$	51%
5	$X_1 X_4$	$D_8 D_{21} D_{28}$	5	24	39.81	$7.46633\text{E}-11$	42%
6	$X_1 X_4$	$D_8 D_{17} D_{21} D_{28}$	6	23	39.05	$6.19539\text{E}-11$	50%
7	$X_1 X_4$	$D_8 D_{17} D_{21} D_{25} D_{28}$	7	22	41.50	$3.10194\text{E}-11$	100%
8	$X_1 X_4$	$D_8 D_{17} D_{21} D_{23} D_{25} D_{28}$	8	21	41.49	$3.71087\text{E}-11$	84%
FULL	$X_1 X_2 X_3 X_4$		4	25	36.30	$5.51451\text{E}-10$	7%

Table 5: Candidate models for the Scottish Hill Data

Model	Predictors	Dummies	df1	df2	F	Quantability	RTOI
1	X_1		1	33	180.87	$6.0476E - 15$	$< 1\%$
2	X_1	D_7	2	32	187.05	$2.21014E - 18$	$< 1\%$
3	$X_1 X_2$	D_{18}	3	31	356.10	$4.27502E - 24$	$< 1\%$
4	$X_1 X_2$	$D_7 D_{18}$	4	30	573.90	$7.20579E - 28$	1%
5	$X_1 X_2$	$D_7 D_{18} D_{33}$	5	29	767.53	$7.17711E - 30$	100%
6	$X_1 X_2$	$D_7 D_{18} D_{19} D_{33}$	6	28	760.36	$1.17176E - 29$	61%
7	$X_1 X_2$	$D_6 D_7 D_{18} D_{19} D_{31}$	7	27	767.57	$2.32597E - 29$	31%
FULL	$X_1 X_2$		2	32	181.66	$3.39797E - 18$	$< 1\%$

Again the optimal model identifies the outliers proposed by literature. The emphasis here is on how effective the proposed method is in the case where outliers are masked.

6.5 Moore Data

This data set was presented in Weisberg (1981) to illustrate variable selection. Here it is used for the same purpose. The dependent variable is the oxygen uptake from five chemical measurements (the predictors). There are 20 observations. The historical view of the “best” subset model is the one that is based on X_3 and X_5 , while observations 1, 7, 15, 17 and 20 are identified as “influential”, either individually or as a group (Chatterjee and Hadi, 1986).

The optimal model based on quantabilities deviates from the suggested model and outliers. The model based only on X_3 is proposed and fewer observations are identified as being outliers. Observations 1 and 20 are obvious outliers, but observations 4 and 6 only seem to be jointly outlying with these two. This view differs from that of the traditional one-at-a-time procedures.

Table 6: Candidate models for the Moore Data based on Quantabilities

Model	Predictors	Dummies	df1	df2	F	Quantability	RTOI
1	X_3		1	18	41.61	5.45393E-06	< 1%
2	X_3	D_1	2	17	61.57	1.63236E-08	9%
3	X_3	$D_1 D_{20}$	3	16	56.97	9.25014E-09	16%
4	X_3	$D_1 D_4 D_{20}$	4	15	61.02	4.23384E-09	35%
5	X_3	$D_1 D_4 D_6 D_{20}$	5	14	73.70	1.49282E-09	100%
6	X_3	$D_1 D_4 D_6 D_9 D_{20}$	6	13	79.59	1.71542E-09	87%
7	X_3	$D_1 D_4 D_6 D_9 D_{16} D_{20}$	7	12	84.43	2.96584E-09	50%
FULL	$X_1 X_2 X_3 X_4 X_5$		5	14	20.70	5.04422E-06	< 1%

6.6 Rousseeuw Artificial Data

Rousseeuw and Leroy (1987) used an artificial data set to illustrate the *exact fit property* of robust regression. They state that if the “majority of the data follow a linear relationship *exactly*, then a robust regression should yield this equation”. Such a regression technique is then said to have the exact fit property. Rousseeuw and Leroy created a data set of 25 observations of which the first 20 satisfy the equation

$$y = x_1 + 2x_2 + 3x_3 + 4x_4 \quad (20)$$

and the last 5 observations fall outside this hyperplane. A least squares fit (without an intercept) to the data is unable to find it, although 80% of the observations lie on the same hyperplane. Here we have an illustration of the phenomenon of *swamping*, where a few outlying observations can let inliers appear to be outlying. This is indeed the case here, since some of the clean observations appear to be more outlying and influential than some of the outliers. Using the holistic regression model and variable selection with quantabilities, the model (21) is readily found, as illustrated by table 7.

This example illustrates the success of the proposed method in the presence of mild swamping.

Table 7: Candidate models for the Rousseeuw Data based on Quantabilities

Model	Predictors	Dummies	df1	df2	F	Quantability	RTOI
1	X_4		1	23	40.55	1.69159E-06	< 1%
2	$X_3 X_4$		2	22	48.56	8.53367E-09	< 1%
3	$X_2 X_3 X_4$		3	21	114.32	3.61974E-13	< 1%
4	$X_2 X_3 X_4$	D_{23}	4	20	136.64	3.19871E-14	< 1%
5	$X_2 X_3 X_4$	$D_{23} D_{24}$	5	19	252.96	1.07619E-16	< 1%
6	$X_1 X_2 X_3 X_4$	$D_{23} D_{24}$	6	18	358.45	1.01287E-17	< 1%
7	$X_1 X_2 X_3 X_4$	$D_{23} D_{24} D_{25}$	7	17	711.86	1.04047E-19	< 1%
8	$X_1 X_2 X_3 X_4$	$D_{21} D_{23} D_{24} D_{25}$	8	16	1050.63	2.78811E-20	< 1%
9	$X_1 X_2 X_3 X_4$	$D_{21} D_{22} D_{23} D_{24} D_{25}$	9	15	∞	0	100%
FULL	$X_1 X_2 X_3 X_4$		4	20	93.04	1.24876E-12	< 1%

7. Conclusion

A method is proposed to essentially identify observations influential on the fit and at the same time to do predictor subset selection. Quantabilities are used as a useful index (the RTOI) to identify optimal subsets. This is not possible using the ordinary available selection criteria.

Various data sets are used to illustrate the capability of the proposed technique to identify outliers and influential observations (individually and jointly) under trying conditions, e.g. masking and swamping. Furthermore it is shown that outliers identified in this way can differ from those identified by the traditional one-at-a-time procedures.

The proposed method employs basic tools available in most statistical software packages and is easy to implement. It is most efficient when based on all possible subset fits, which might be a drawback (in terms of computational time) for larger data sets.

There is scope for additional research with regard to the ideas proposed in this paper. A simulation study could give deeper insight into the predictive capabilities of the selected (optimal) model. Furthermore the potential of using quantabilities in some way as a general model selection criterion can be investigated. The author has done some work on this problem.

A referee suggested exploring the proposed technique on simulated data when the usual linearity, homoscedasticity and normality assumptions are not met. The possibility of extending the method from linear models to generalised linear models is also mentioned. These valuable suggestions will be used in further theoretical developments and research of the method.

A second referee pointed out that standard graphical techniques as additional tool in making the final decision should be mentioned. This is a

valid point. Plots such as partial residual plots and partial regression plots can be very useful in this regard.

Computations were done using SAS V8 and Microsoft Excel 2000.

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